

Phylogenetic Analysis of Tubulin Protein Sequences

Methods:

1.1 Dataset design and sequence retrieval

Protein sequences representing the tubulin family were downloaded from the UniProt Knowledgebase (UniProtKB). The main objective was the comparison of complete sequences of α , β , and γ -tubulins from a range of diverse eukaryotes. The final data set, after data curation, consisted of 10 α -, 10 β - and 10 γ -tubulin sequences from each of five different lineages covering the breadth of the eukaryotic tree of life: animals, fungi, plants or green algae, amoebozoa and kinetoplastids. The final curated dataset contained 30 protein sequences.

1.2 Sequence curation

The sequences were selected based on tubulin type, organism, annotation, and sequence length (full-length or near full-length) and reviewed UniProtKB entries were used when available. Sequences that were not themselves tubulins, but rather proteins associated with tubulin function, as well as unrelated proteins (e.g. a transcription initiation factor) were not considered. Removed entries included kinesin-like proteins, alpha-tubulin N-acetyltransferases, gamma-tubulin complex subunits, spindle pole body components, and unrelated proteins such as transcription initiation factors.

1.3 Handling of paralogs and inconsistent entries

For each organism or organism group, one representative sequence was selected for each tubulin type. This ensures reduction in overrepresentation from species with multiple tubulin paralogs and it also helps in maintaining a balanced dataset across alpha-, beta-, and gamma-tubulin groups. The *Trypanosoma* sequences were labelled as *Trypanosoma_brucei_complex* because exact-species reviewed entries were limited. Even though the gamma-tubulin entry from this group was unreviewed it was retained because of its clear gamma-tubulin annotation and appropriate sequence length.

1.4 Domain-related considerations

No other domain extraction was performed. The selected UniProt entries were already annotated as alpha-, beta- or gamma-tubulin proteins and sequence lengths compatible with expected full-length tubulin proteins. Only full-length or near full-length tubulins were retained. Tubulin associated proteins, tubulin modifying enzymes and tubulin complex subunits were not included as they were not the target protein family members for this analysis.

1.5 Multiple sequence alignment

The curated protein sequences were aligned using MAFFT. The alignment was performed with the `--localpair` and `--maxiterate 1000` options, using a local pairwise alignment strategy with

iterative refinement. The --reorder option was used to reorder sequences in the alignment output. The raw multiple sequence alignment contained 525 columns and was saved as tubulin_raw_msa.fasta.

1.6 Alignment trimming

TrimAl tool was used with the -automated1 option to trim the raw alignment. Before building a phylogenetic tree, trimming was done to eliminate areas that were poorly aligned and had a lot of gaps. The -fasta option was used to save the output in FASTA format. Phylogenetic inference was performed using the final trimmed alignment, which had 365 columns.

1.7 Phylogenetic tree construction

IQ-TREE tool was used to generate the phylogenetic tree, and -st AA was used to specify the amino acid sequence type. ModelFinder's -m MFP option was used to choose models automatically. LG+I+G4 was determined by IQ-TREE to be the best-fit model. 1000 ultrafast bootstrap repetitions with -B 1000 and 1000 SH-aLRT replicates with -alrt 1000 were used to estimate branch support.

1.8 Tree visualization and annotation

iTOL online tool was used to view and interpret the phylogenetic tree. The tree is rooted for clearer interpretation, and node support values are displayed on internal branches. Alpha-, beta-, and gamma-tubulin sequences were marked using a color-strip annotation. After being exported from iTOL, the final figure was used for biological interpretation. Instead of focusing on detailed lineage-specific tubulin evolution, the annotated tree was created to highlight broad paralog-level links among representative eukaryotic tubulin sequences.

2. Discussion

2.1 Overall phylogenetic pattern

The maximum-likelihood tree roughly divided the curated tubulin sequences into three primary groups, which corresponded to alpha-, beta-, and gamma-tubulins. The idea that these tubulin types are different paralogous groupings within the tubulin protein family is supported by this. Alpha-, beta-, and gamma-tubulin lineages appear to be distinct evolutionary groups within eukaryotes, since the primary clustering pattern was based more on tubulin type than organism identity.

2.2 Biological interpretation of tubulin groups

In the tree, alpha- and beta-tubulins formed distinct but connected groupings. Because alpha- and beta-tubulin are related proteins that work together as the fundamental heterodimeric building block of microtubules, this has biological significance. Given its unique function in microtubule nucleation, gamma-tubulin formed a distinct category from alpha- and beta-tubulins. Alpha-,

beta-, and gamma-tubulins are functionally and evolutionarily separate tubulin paralog groupings, according to the tree's overall interpretation.

2.3 Curation choices and limitations

One representative alpha-, beta-, and gamma-tubulin sequence for each organism or group of organisms was intentionally included in the dataset. This kept the comparison balanced and decreased the overrepresentation of species with many paralogs. It did, however, also limit the resolution of tubulin evolution specific to lineages. It is important to be cautious when interpreting certain internal branches that have moderate or poor support. Due to the lack of exact-species reviewed entries and the unreviewed gamma-tubulin entry, the *Trypanosoma* entries were handled with caution.

2.4 Prokaryotic tubulin-like proteins and future directions

Since this analysis concentrated on eukaryotic alpha-, beta-, and gamma-tubulins, prokaryotic tubulin-like proteins were taken into consideration but excluded from the main tree. The analysis would change from a narrow comparison of eukaryotic tubulin to a more comprehensive investigation of the FtsZ/tubulin-superfamily if homologs like FtsZ or BtubA/B were included. Such an expanded analysis would require wider prokaryotic sampling, careful homolog selection, and possibly domain-level comparison. These proteins may be used in future research to investigate more intricate evolutionary connections within the tubulin superfamily.

3. Phylogenetic Tree

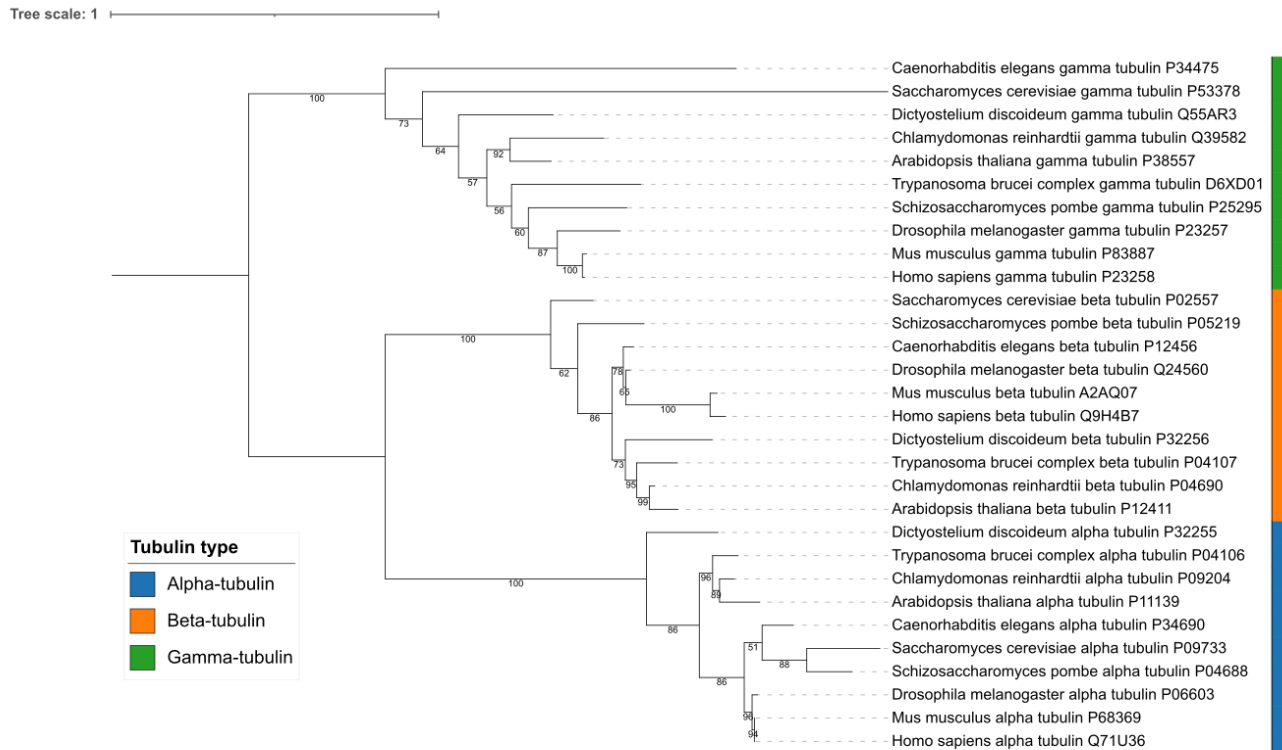


Figure 1. Rooted and annotated maximum-likelihood phylogeny of representative eukaryotic alpha-, beta-, and gamma-tubulins.

The tree was constructed using IQ-TREE from a trimAl-trimmed MAFFT alignment and visualized in iTOL. The tree node support values are displayed on internal branches. Colored strips indicate tubulin type. The tree shows broad separation of alpha-, beta-, and gamma-tubulin sequences into distinct paralog groups. Branch lengths are shown according to the tree scale; detailed support values are provided in the IQ-TREE output files as well.